

BEC503 Digital Communication — Assignment 3 — Entropy, source coding and channel capacity

Module 3: Information Theory

Field	Value
Maximum marks	20
Weightage	10% of CIE
Issue date	< <TBD> >
Due date	< <TBD> >
Items	8 problems + 3 theory questions
CO / Bloom coverage	CO3: 11 · L1–L2 0, L3–L4 6, L5–L6 5

Objective

Compute the two fundamental limits — the entropy floor for representing a source and the capacity ceiling for a channel — and locate real schemes relative to both.

Learning outcomes

- Student will be able to compute entropy, code efficiency and redundancy for a given source.
- Student will be able to construct Huffman and Lempel-Ziv codes and compare them on the right criteria.
- Student will be able to compute mutual information and capacity for discrete channels.
- Student will be able to apply the Information Capacity Law and locate a system relative to the Shannon limit.

Part A — Problems (17 marks)

P1. A source emits six symbols with probabilities 0.30, 0.25, 0.20, 0.12, 0.08, 0.05. Compute the entropy, construct a Huffman code, compute the average length, efficiency and the variance of the code length. Then construct a second Huffman code using a different tie-breaking rule and compare the variances.

[C03 | L4 | 3 marks]

P2. For the binary source with $p = 0.85$, compute the entropy, then construct Huffman codes for the first, second and third extensions. Tabulate the average length per source symbol at each n and plot the convergence towards the entropy. [C03 | L4 | 3 marks]

P3. Apply the Lempel-Ziv algorithm to the binary sequence 000101110010101110100. Show the dictionary as it builds, give the encoded output, and state the compression achieved. Explain why the result is poor on such a short input. [C03 | L4 | 3 marks]

P4. For a binary symmetric channel with $p = 0.15$ and inputs with $p(0) = 0.7$, compute $H(X)$, $H(Y)$, $H(Y|X)$, $H(X|Y)$ and $I(X;Y)$. Then find the input distribution that maximises I and state the capacity. [C03 | L4 | 3 marks]

- P5.** Plot the capacity of the BSC and the BEC on the same axes against disturbance probability from 0 to 0.5. Identify the probability at which the BEC carries exactly twice the BSC and explain the shape of each curve at its endpoints. [C03 | L5 | 2 marks]
- P6.** A satellite link has 30 MHz of bandwidth and a received SNR of 12 dB. Compute the capacity. Then determine what bandwidth would be needed to carry the same rate at an SNR of 6 dB, and comment on which resource is cheaper to buy. [C03 | L5 | 1 marks]
- P7.** Locate uncoded BPSK, uncoded 16-QAM and a rate-1/2 coded QPSK system on the spectral-efficiency against E_b/N_0 plane, with the capacity boundary drawn. State the coding gain still available to each. [C03 | L5 | 1 marks]
- P8.** Explain, using the data processing inequality, why a receiver that quantises its input to one bit before decoding cannot perform as well as one that keeps soft values. Estimate the loss in decibels. [C03 | L5 | 1 marks]

Part B — Theory (3 marks)

- T1.** The channel coding theorem was proved in 1948 but capacity-approaching codes did not appear until 1993. Explain what makes a non-constructive existence proof so hard to turn into a working code. [C03 | L4 | 1 marks]
- T2.** Argue whether entropy is the right measure of information for a source whose symbols have different costs to transmit, and propose how the theory would need to change. [C03 | L5 | 1 marks]
- T3.** A binary symmetric channel with $p = 0.9$ is better than one with $p = 0.5$. Explain why, and state what that says about the shape of the capacity curve. [C03 | L4 | 1 marks]

Submission guidelines

- Format: PDF, single file
- File name: <USN>_BEC503_Assignment3.pdf
- Channel: LMS upload
- Late policy: 10% per day, nothing after 5 days
- Every numerical answer must carry units and a stated reference direction.

Assessment rubric

Criterion	Weight	Excellent	Good	Satisfactory	Needs improvement
Correctness of entropy and capacity calculations	8	All quantities computed correctly with intermediate terms shown.	Correct results with some intermediate steps omitted.	Method sound but several arithmetic errors.	Method or results substantially wrong.
Code construction and verification	5	Codes constructed correctly and verified against Kraft and the entropy bound.	Codes correct but unverified.	Codes constructed with prefix or length errors.	Codes invalid or absent.

Criterion	Weight	Excellent	Good	Satisfactory	Needs improvement
Plots and comparative analysis	4	Curves correctly shaped, endpoints justified, comparisons quantified.	Plots correct but comparisons qualitative only.	Plots present with shape or scale errors.	Plots missing or wrong.
Interpretation of the fundamental limits	3	Explains what each limit forbids and where each assumption applies.	Some interpretation offered.	Limits quoted without interpretation.	No interpretation attempted.
Total	20				

Course outcome attainment

Course outcome	Marks carried
CO3	18
CO4	2
Total	20

Academic integrity

Individual submission. Identical submissions score zero for both parties.